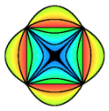
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Semiconductor
Physics Group

Semiconductor Physics Lab Course

Optoelectronic Devices

B 7: Laser Diodes

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1. Introduction

The semiconductor laser is a relatively young result of fundamentally oriented and applied research. Only in 1960, the first gas laser, that made use of atomic transitions in order to produce intense, coherent and monochromatic radiation, was built. Gas lasers are thermally extremely stable and the emitted radiation has very small divergence angles given the long cavity length. Such lasers are commonplace in research and technology.

The first semiconductor lasers on the basis of GaAs diodes were developed already in 1962, shortly after the advent of gas lasers, whereas the production of more efficient laser diodes on the basis of semiconductor double heterostructures took until 1969. In 1970 the first glass fibres for optical fibre communication were produced. With that, the essential components for modern optical communication had been made and a fast development started.

In the past years the share of semiconductor lasers of the total sales volume of the laser market has increased excessively. Today, the applications in telecommunication make the greatest part (69%) of the laser diode market followed by optical data storage with 22%. All other important applications only have market shares of 1…3%.
Nowadays, also high performance lasers are made from semiconductor materials. For material processing they are an energy-efficient alternative to gas lasers.

At present, one strives for laser diodes which have larger efficiencies, improved thermal stability, emission in different wavelength ranges, ultrafast modulation speed, and so on. This variety of requirements leads to specialized and complex devices. A current trend is, for example, the application of quantum effects to increase the efficiency, stability and speed. Semiconductor systems with carrier confinement such as quantum wells, quantum wires and quantum dots (systems of 2, 1 and 0 dimensions) are explored and optimized as laser gain medium.

2. Principle of Action

2.1. What is a LASER?

A semiconductor laser uses the radiative transitions of electrons between the valence and conduction band. Such transitions have to be energy and momentum conserving, at least at low carrier density. Because of the negligible photon momentum they are practically vertical in k-space. One distinguishes three basic types of interaction of photons with semiconductors (or matter in general): (a) spontaneous emission, (b) absorption and (c) stimulated emission.

(a) Spontaneous emission is the recombination of an exisiting electron hole pair (produced beforehand) without outer cause under emission of a photon. Due to this statistical process the emitted photons have coincidental directions and phase relations, the light is incoherent. The respective line shape is proportional to the product of the joint density of states and Fermi-Dirac distribution. For (isotropic) parabolic bands and high temperatures or very low charge carrier densities (Boltzmann approximation) it is approximately given by:

$$I_{sp}(\hbar\omega) \propto \omega^2 \sqrt{\hbar\omega - E_g}\, \exp\left(-\left[\hbar\omega - E_g\right]/kT\right).$$

(b) An incoming photon of sufficiently high energy (larger than or equal the band gap) can

be absorbed and generates an electron-hole pair. The line shape of absorption for parabolic bands reflects the square-root dependence of the joint density of states:

$$I_{12}(\hbar\omega) \propto \sqrt{\hbar\omega - E_g}\,.$$

(c) An incoming photon stimulates the recombination of an electron-hole pair. At this, a further photon is emitted, which has the same direction and phase as the first one. The resulting radiation is coherent. This process is used for light amplification in a LASER (*l*ight *a*mplification by *s*timulated *e*mission of *r*adiation).

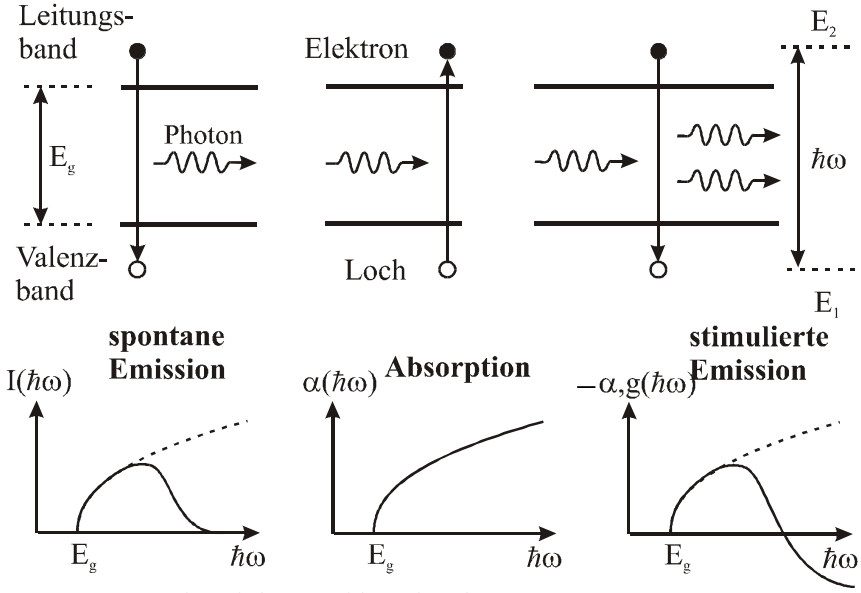


Fig. 1 Interaction of photons with semiconductors

In thermodynamic equilibrium the transition rates of absorption and emission R_{12}, R_{sp} and R_{21} fulfil the rate equation $R_{12} = R_{21}+R_{sp}$.

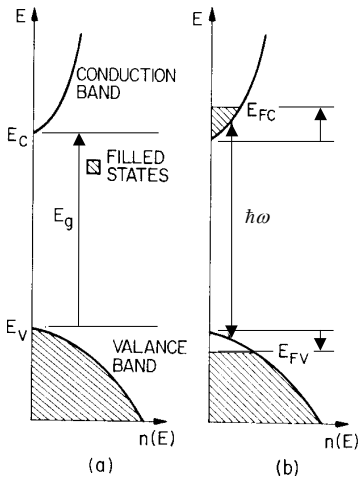


Fig. 2 Occupation of the states in the valence and conduction band for T = 0K (a) in equilibrium and (b) at inversion (from [Sze])

A working laser diode is a forward biased p-n-junction. At this, charge carriers are injected into the depletion layer, which leads to a non-equilibrium system. The quotient of the transition rates R_{12} and R_{21} of absorption und stimulated emission at an optical wavelength $\hbar\omega$ and the temperature T is given by:

$$\frac{R_{12}}{R_{21}} = \exp\left[\frac{(\hbar\omega - E_g) - (E_{Fc} - E_{Fv})}{kT}\right].$$

Important for the amplification of the optical wave is the position of the quasi-Fermi-levels E_{Fc} and E_{Fv}. In equilibrium the quotient is always larger than 0 and the absorption rate is larger.

If the difference E_{Fc}-E_{Fv} is larger than the difference between the energy of the optical wave and the band gap energy, the wave is amplified. This condition is equivalent to the thermodynamical condition of inversion.

For statistical reasons, the line shape of stimulated emission for the case of isotropic parabolic bands can be considered proportional to the product of the joint density of states and the difference of the occupation probabilities of the valence and conduction band:

$$I_{21}(\hbar\omega) \propto \sqrt{\hbar\omega - E_g}\,(f_v - f_c).$$

The carrier injection has to be sufficiently large to conserve this non-equilibrium distribution. From the line shape of the gain spectrum on the right hand side in Fig. 1 one finds that there is a direct relation between the highest possible amplification (solid line), absorption coefficient (dashed line) and occupation. Therefore, only an active material with high absorption offers large gain and subsequently high amplification.

The ratio of stimulated and spontaneous emission rates is directly proportional to the photon density $\rho(\hbar\omega)$:

$$\frac{R_{21}}{R_{sp}} \propto \frac{\rho(\hbar\omega)}{(\hbar\omega)^2}.$$

This means, in order to suppress spontaneous emission as much as possible, the photon density in the active region has to be large. In practical devices, this is mostly realized by a layer structure for wave guiding and the presence of a resonator, the so-called cavity, providing feedback. This leads to an increase of photon density in the active region at wavelengths that coincide with resonator modes. Therefore, a LASER is an optical oscillator consisting of an amplifying medium and the resonator giving feedback on certain resonance modes.

2.2. Semiconductor Materials for LASERs

Many semiconductor materials with direct band gaps are suitable for lasers operating in various spectral ranges. Most important for semiconductor lasers are, however, III-V-compounds such as (Al,Ga,In)N, $Al_xGa_{1-x}As_ySb_{1-y}$ and $Ga_xIn_{1-x}As_yP_{1-y}$. For example, the system $Ga_xIn_{1-x}As_yP_{1-y}$ on InP-substrate plays an important role in optical communication for a wavelength of 1.55 μm due to its radiating properties and the good lattice match.

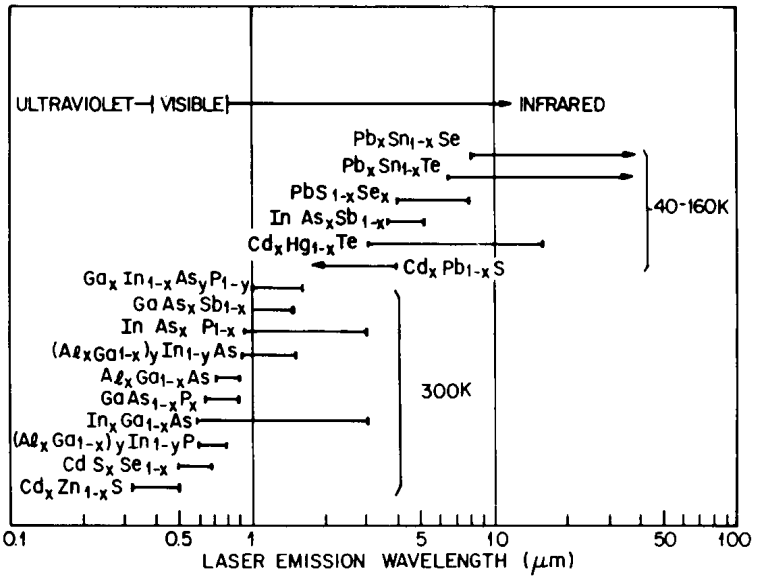


Fig. 3 Semiconductor materials for various wavelength [Sze]

An important quantity for active media in a laser is the gain g (negative absorption g=-α), $[g]=cm^{-1}$. It gives the intensity increase of the wave during passing the way z in the medium by the exponential relation (like the Lambert-Beer law of absorption)

$$I(z) = I_0 \exp(+g\,z)$$

Low dimensional systems with quantum wells (2D), quantum wires (1D) and quantum dots (0D) in the active region can be made by embedding a low-gap material into a material with higher band gap energy.

The restriction of the motion of the charge carriers in one more directions leads to modified densities of states and different transition probabilities. The square root-like shape of the density of states for band-band transitions in the bulk semiconductor becomes a step function in the quantum well (one step for each sub-band) and even discrete lines for each state in the quantum dot (see Fig.4).

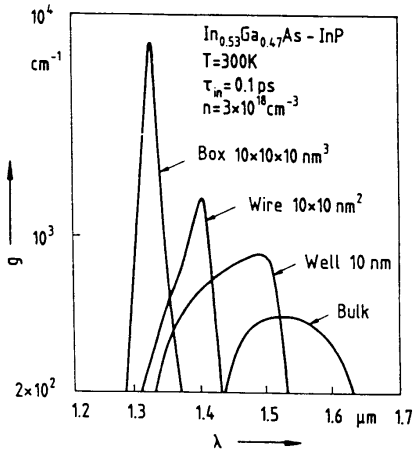


Fig. 4 Increase of maximum gain g for low-dimension systems of InGaAs on InP [Sze]

The possible gain for stimulated emission can be remarkably larger in low-dimensional systems. For practical use of the theoretically high gain in quantum dot systems it is necessary to fabricate a high density of equal quantum dots in the whole active region, in order to superimpose the gain curves of the single quantum dots in the best way.

If these conditions are fulfilled, for quantum dot lasers the lowest threshold currents are expected.

2.3. Active Region – Electrical Optimization

A widely used laser structure is the double heterostructure (DHS). The active material (e.g. GaAs) is embedded between two layers of a material with a higher band gap (e.g. AlGaAs). The outer layers are highly p- and n-doped, respectively. The band gap steps at the heterojunctions form potential wells for the minority charge carriers that can be collected in the active region in this way.

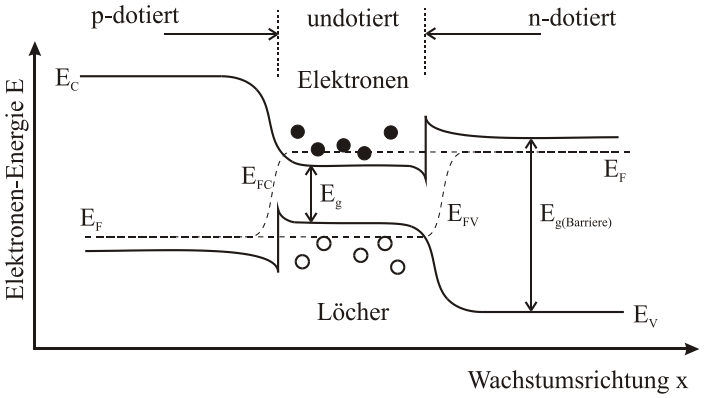


Fig. 5 Double heterostructure as p-i-n transition in forward direction: The solid lines are the valence and conduction band edge E_V and E_C, respectively. The Fermi energy E_F splits into the quasi-Fermi levels E_{FC} and E_{FV} in the active region. There you have inversion, the quasi-Fermi levels lie in the bands.

The discontinuity of the conduction band and the bending of the bands at the hetero junction give the barrier height. For a barrier height of about E_B=0.3 eV and a total electron density of n=2x10^{18}cm^{-3} at room temperature one finds $\hat{n}$=3.7x10^{14} cm^{-3} electrons at energies above E_B. This part of the electrons is lost for the radiative recombination in GaAs with high probability and contributes to the losses in the device in general (leakage current).

2.4. Adjoining layers – Optimization of Wave Guiding

Because of the different refractive indexes of the materials, the layer order in a DHS leads to a wave guiding in the active layer (we are lucky here because a higher band gap leads to lower refractive index for a given energy, why!?). For example, in the AlGaAs/GaAs system, the refractive index is inversely proportional to the band gap energy. This means, that a structure of a GaAs layer between two AlGaAs layers is like a glass fiber in which light can be guided (Fig. 6).

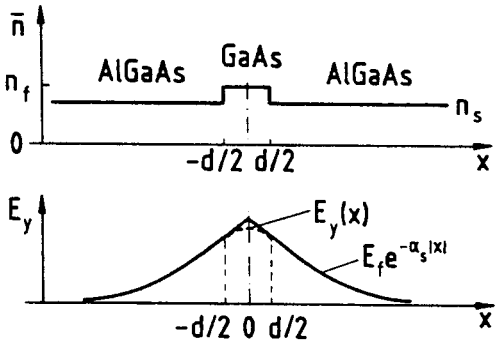


Fig. 6 Profile of refractive indexes of a AlGaAs/GaAs double heterostructure and approximate field distribution of the optical wave [Ebeling]

For the behavior of the laser the ratio of the light intensity in the waveguide to the total intensity inside and outside of the waveguiding layer is of interest. It is called the optical confinement factor Γ.

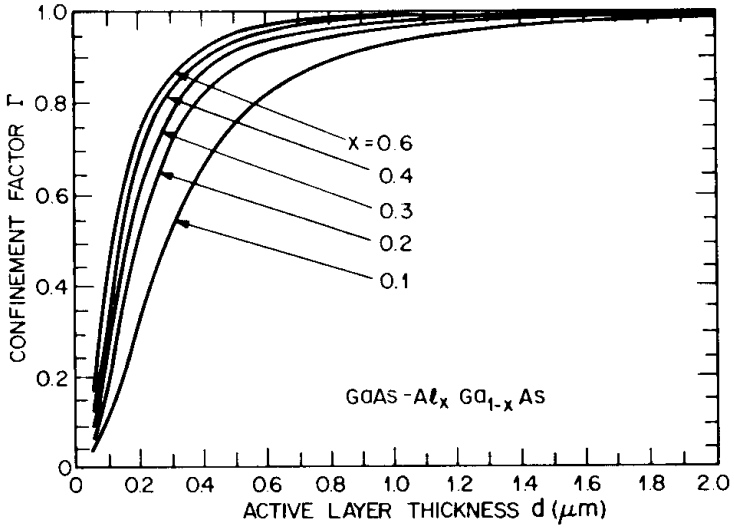


Fig. 7 Optical confinement factor of the principle mode in a AlGaAs/GaAs double heterostructure vs. GaAs layer thickness and barrier composition [Sze]

In quantum well lasers, the active layers are too thin to provide an effective wave guiding by themselves. Therefore, a so-called SCHS (*s*eparate *c*onfinement *heterostructure*), in which active layer and waveguide are different layers, is devised. The active layer with one or more quantum wells (Why is it not a good idea to use too many QWs?) is embedded in a material of higher band gap. The layer thicknesses of the outer waveguide and the profile of the refractive index can be adapted optimally to the requirements.

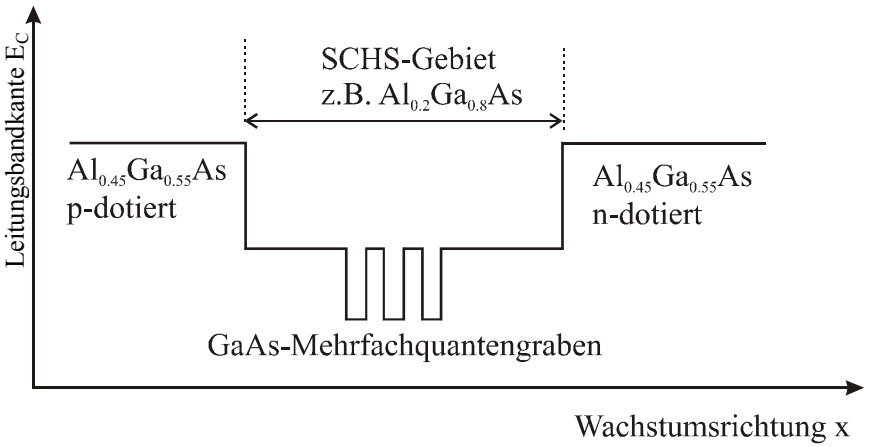


Fig. 8 Example for a SCHS: Functional separation of active layer (GaAs multiple quantum well) and waveguide layer (here $Al_{0.2}Ga_{0.8}As$).

2.5. Concentration of Carriers and Light Wave Due to Lateral Structuring

Till now one started out from the assumption, that the laser emits light over its whole width, that means one considered the injection layer and the waveguide as being laterally infinite. Current injection is only desirable in a narrow stripe, in order to control a high current density in the active layer at low external currents.

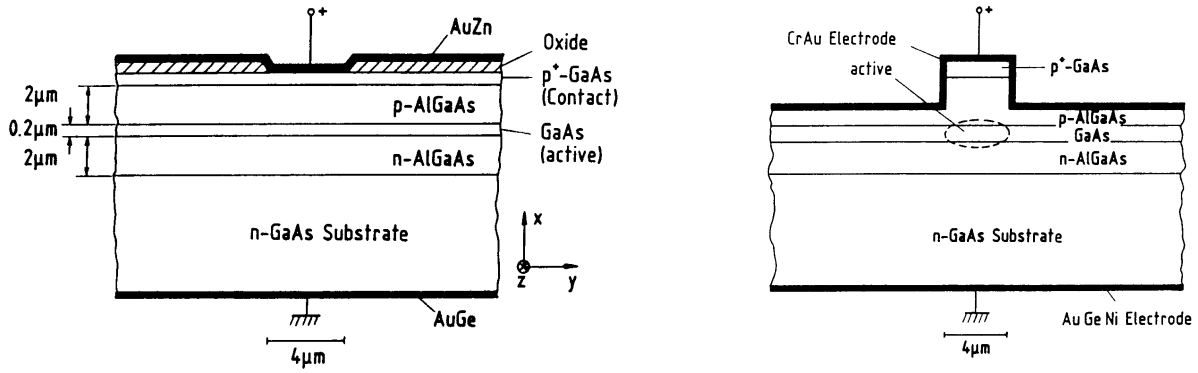


Fig. 9 Cross section of an oxid stripe laser (left hand side) as an example for „gain guiding“ and a ridge waveguide laser (right hand side) as an example for „index guiding“ [Ebeling]

A technologically simple construction following this principle is the so-called oxide stripe laser (Fig.9, left hand side). Its surface contact is isolated by an oxide layer except for a narrow, long stripe parallel to the cavity. The lateral extension of the light wave is determined by the current density profile, that determines the amplification in the material. This is called "gain guiding". Besides of the confinement of the current path it is useful to guide the light wave laterally to the stripe, which is called "index guiding".

A simple example for an index guided laser is the so-called ridge waveguide structure (Fig. 9, right hand side). The p-type layer is etched except for a small stripe (mesa), and the whole surface is metallized. The different thicknesses of the p-AlGaAs lead to a change of the effective refractive index, such that the wave is guided below the mesa. Besides, this geometry leads to a more effective confinement of the current path. Practically lasers exhibit a mixture of gain and index guiding.

2.6. Fabry-Perot Resonator

Besides of the amplification in the active layer and a good wave guiding, for a laser one needs a resonator. It consists of plane-parallel mirrors between which self-stimulating oscillations may occur.

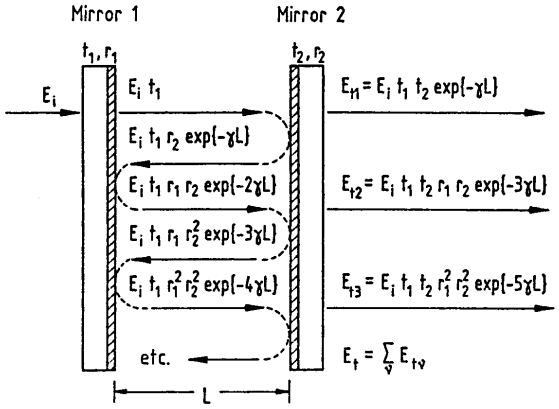


Fig. 10 Fabry-Perot Resonator and superposition of waves [Ebeling]

Usually, for semiconductor lasers Fabry-Perot resonators are used. In the simplest case crystallographically smooth cleaved facets at both ends of the active stripe build the mirrors. The reflectivities only depend on the refractive index of the material. Usual values for the reflectivity of an AlGaAs structure are 30…33%. In the active region, the wave is guided in different materials, such that one could find an effective refractive index $\bar{n} = \bar{n}(\lambda)$. In order to achieve amplification, the resonator has to have the length of a multiple of the wavelength. Therefore, only waves with wavelength $m\lambda = 2L\bar{n} \quad with \quad m = 1,2,...$ are amplified. The length of the resonator is L. With that, the mode separation $\Delta\lambda$ for large m is:

$$\Delta\lambda = \frac{\lambda^2 \Delta m}{2\bar{n}L\left[1 - (\lambda/\bar{n})(d\bar{n}/d\lambda)\right]}$$

The bracket describes the dispersion of the refractive index $\bar{n} = \bar{n}(\lambda)$ and is nearly 1 in gas lasers. Because of the short resonator lengths of semiconductor lasers (several 100 µm) the mode separation is much larger than for gas lasers.

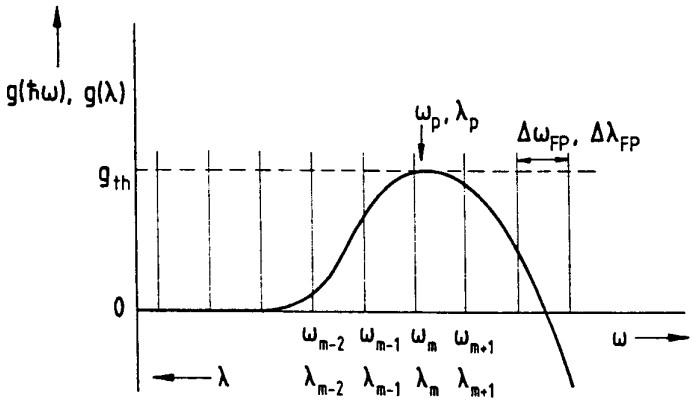


Fig. 11 Wavelength dependence of the gain and position of the Fabry-Perot resonances ω_m, λ_m (schematic). The peak gain occurs at ω_p and λ_p.

3. Characterization of Laser Diodes

3.1. Electrical Parameters of Laser Diodes

The examination of the current-voltage characteristics is a common method for the characterization of semiconductor devices. In the case of the laser it is used to check the quality of the contacts and determine the size of the ohmic bulk resistances. The I-U characteristic is found by presetting the voltage of the device and measuring the current flowing. The ideal of such a characteristic is the following: Under reverse bias the current I_0 is negligible, and under forward bias it increases exponentially with $I = I_0 \exp[(eU_a/kT)-1]$. U_a is the voltage, and e the electron charge. The real relation mostly is more complicated.

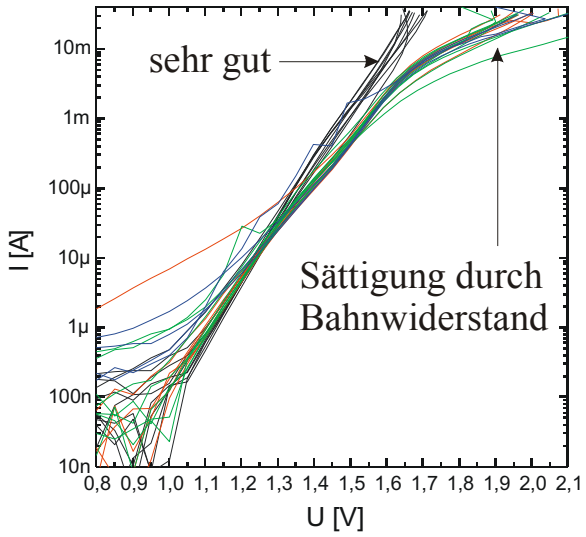


Fig. 12 *I-U* characteristics of real diodes: There are two families of characteristics with different quality. The deviation from the exponential curve for high currents is due to series or bulk resistances in devices.

3.2. Optical Parameters of Laser Diodes

In order to determine important parameters, one records the output light power-current (P-I) characteristic (Fig.13). The laser diode is operated in forward direction and a given (constant or pulsed) current is flowing. The emitted light for each current is collected and detected spectrally integrated. Up to the threshold current I_{th} it only shows a small gradient. In this regime, the emitted light stems from spontaneous and stimulated emission without laser activity. Above the laser threshold at I_{th} the curve increases steeply, and its gradient is the differential efficiency $\eta_{diff} = \dfrac{d(P/\hbar\omega)}{d(I/e)}$ of the laser mode. The total efficiency is given by the ratio of emitted photonic power over the electric pumping power.

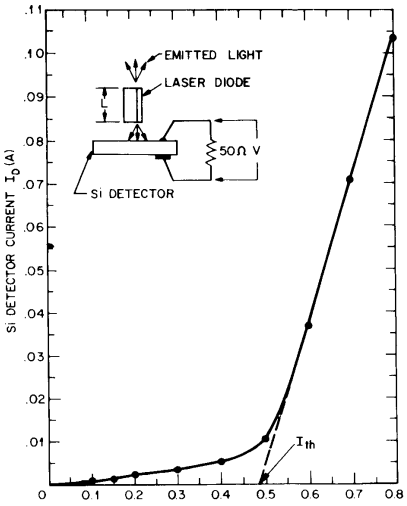


Fig. 13 Typical curve of a P-I characteristic

Knowing the geometry of the active region, from its area and the threshold current one can directly find the threshold current density j_{th}. It is important for comparisons between different structures and for optimization.

Another important relation is the increase of the threshold current with temperature. Around a temperature T_1 it often follows the empirical relation

$$j(T) = j(T_1)\exp\left\{\frac{T-T_1}{T_0}\right\} \propto \exp\left\{\frac{T}{T_0}\right\},\quad T_0 = \left(\frac{d\ln j}{dT}\right)^{-1}.$$

The parameter T_0 has the dimension of a temperature difference and can be given in °C or K. Mostly, T_0 is called characteristic temperature and should be given in Kelvin for reasons of unambiguity. T_0 summarizes the temperature dependent losses and the change of the Fermi-Dirac distribution with temperature. Even an ideal quantum well laser has a finite value of T_0.
At low operating currents, the spectrum of the electroluminescence is that of spontaneous emission, at room temperature a broad line is observed. With increasing current density, the effects of stimulated emission increase and the line shape changes. The line becomes much narrower and the maximum shifts to higher energies. At the threshold current, a spectrum of several sharp lines occurs reflecting the resonator geometry.
Above threshold, under constant operating current, there is a tendency to the formation of only a single very narrow line, the main mode. The higher the operating current (above the threshold), the more the side modes are suppressed, such that for currents high above the threshold there is practically only one mode left. One calls this single mode operation by mode saturation (the details depend on mode coupling and non-linear effects).
Changes in temperature and electron density influence he refractive index as well as the curve of the gain spectrum, which can lead to different modes preferred. This behavior is called mode jumping.

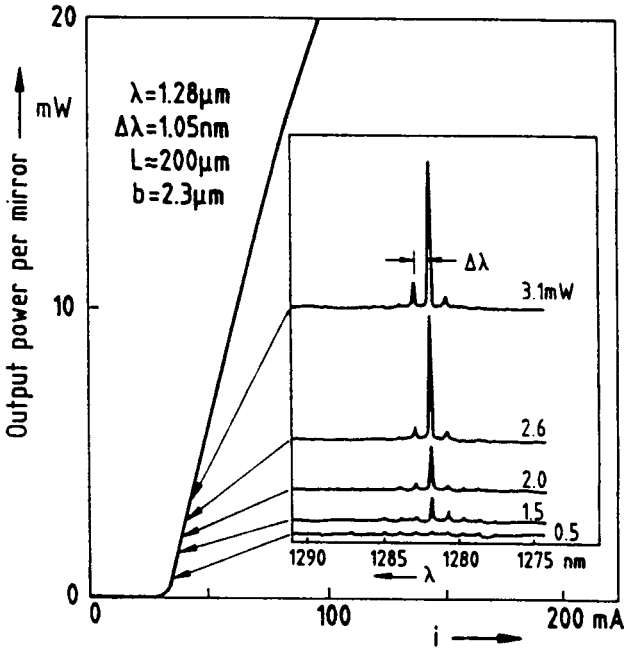


Fig. 14 Typical spectra of Fabry-Perot Lasers at different pumping currents. Increasing mode saturation at high currents. [Ebeling]

4. Literature

[Ebeling] Ebeling, K.J.: *Integrated Optoelectronics*. Berlin - Heidelberg - New York: Springer 1993

[Sze] Sze, S.M.: *Physics of Semiconductor Devices*. 2nd Edition. New York: Wiley&Sons 1981

5. Tasks

(1) Measure the *I-U* and *P-I* characteristics of the given laser diode in forward direction up to 100 mA. Give a statement about the quality of the contacts. Determine the threshold current, the differential efficiency and the total efficiency of the laser diode.
(2) Measure the *P-I* characteristics for pulsed excitation in the range from 0...200 mA at 5 temperatures between 15 and 50°C and determine T_0.
(3) Measure the spectrum of spontaneous emission of the laser diode by operating it at a very low current density as LED. Determine the effective carrier temperature from the gradient in the high-energy range by using the line shape model for spontaneous emission.
(4) Measure the spectrum at a higher operating current below the threshold and again determine the effective temperature. Compare the result with (3) and find conclusions.
(5) Measure the mode spectrum at an operating current above the threshold. Determine the effective refractive index of the active material from the mode separation and the length of the resonator, which has been determined by means of a light microscope before. The dispersion shall be neglected: $(d\bar{n} / d\lambda) \to 0$.
(6) Estimate the power density in the active volume in the laser mode from the given geometric values of the laser.

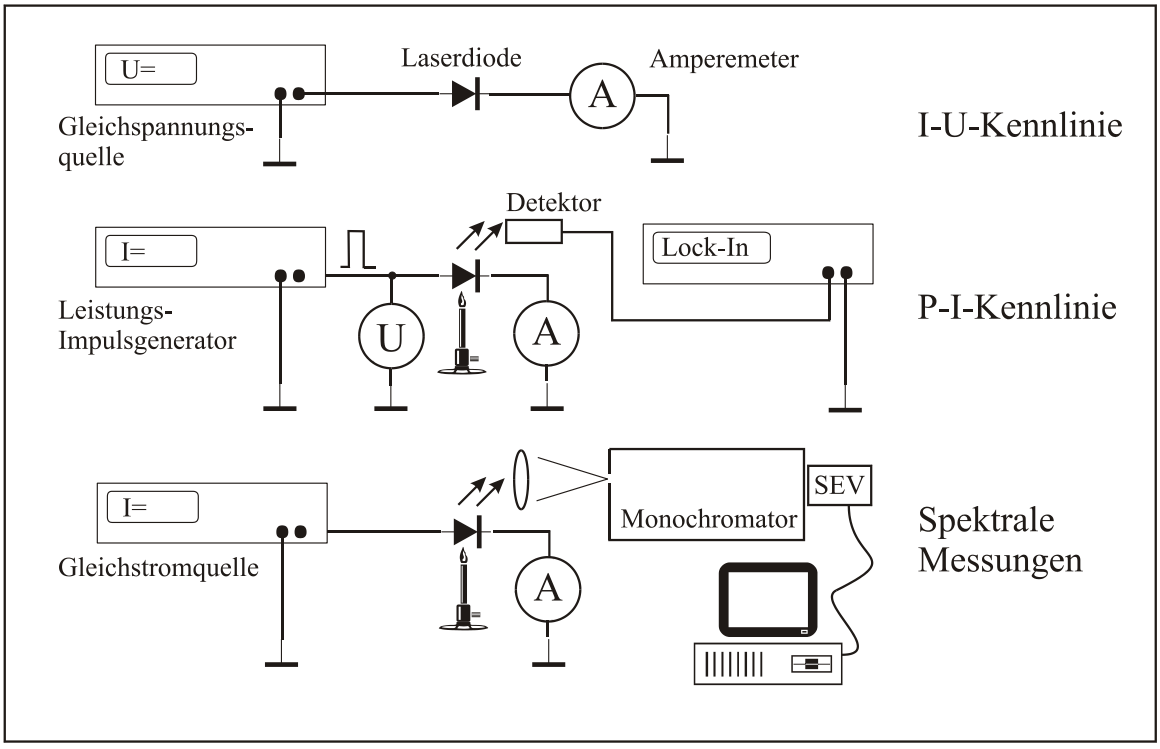


Fig. 15 Principle of the experimental setup. Please use detailed description at the workstation.

Technical note:

In the measurement files the current is saved in A, voltage in V, power in W, wavelength in nm, and intensity in arbitrary units.